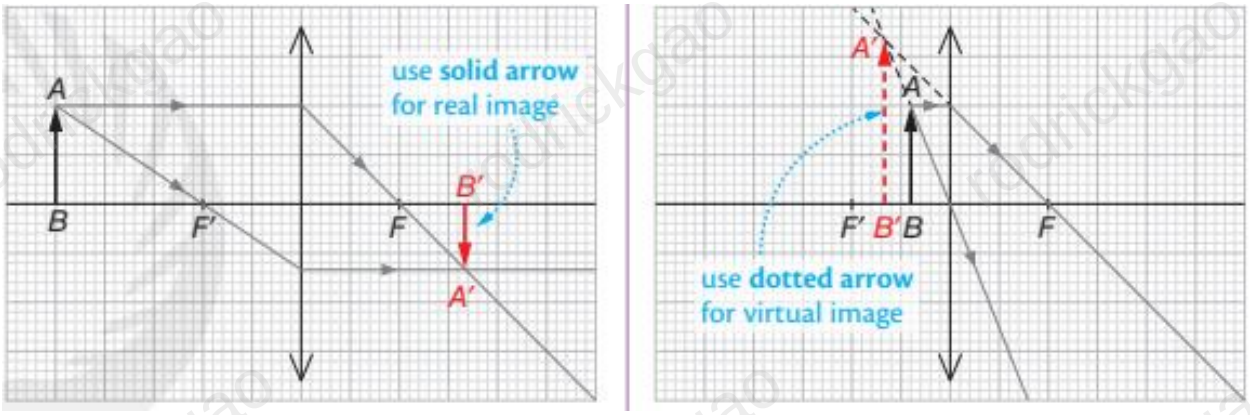
3A03 Lenses

3.3.1 Real and virtual images

A real image is formed by actual convergence of rays and can be captured on a screen. A virtual image is formed where backward ray extensions meet; the rays do not pass through it, so it cannot be directly projected. Virtual images are still visible because light entering the eye appears to originate from their positions.



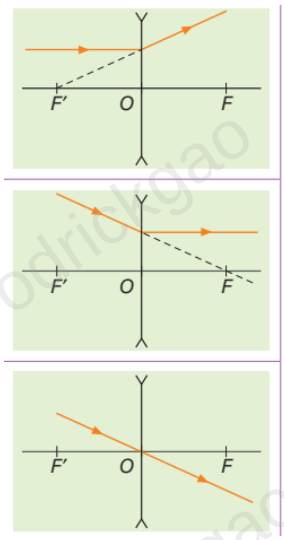
Left: an object beyond the focus forms an inverted real image where rays converge. Right: an object within the focal distance forms an upright virtual image where backward extensions meet.

Textbook sources: 3A03 · PDF 11, 13

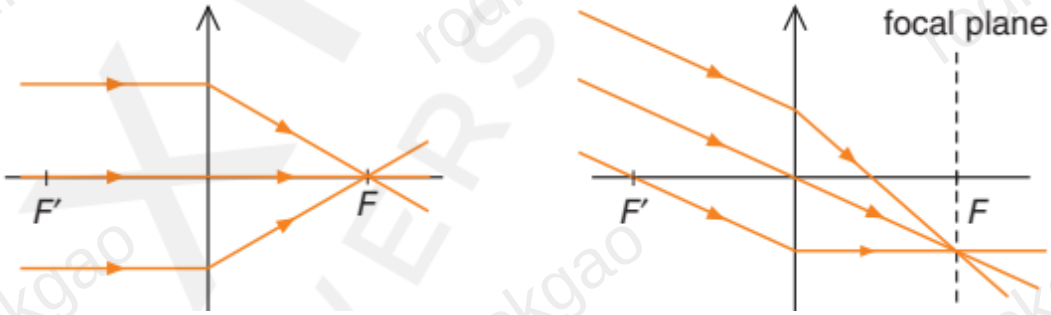
rodrickgao

3.3.2 Concave-lens rays and the focal plane

Three construction rays are useful for a concave lens: an incident ray parallel to the axis emerges as if from the near focus; one aimed at the far focus emerges parallel; one through the optical centre continues straight. For oblique parallel rays entering a convex lens, convergence occurs in the focal plane, but not necessarily at the principal focus.



Three construction rays for a concave lens, showing the axis, focal points and ray directions.



Oblique parallel rays converge in the focal plane after passing through a convex lens.

Textbook sources: 3A03 · PDF 16, 24

3.3.3 · 3.3.4 Real and virtual images and magnification

3.3.3 Image formation by a convex lens

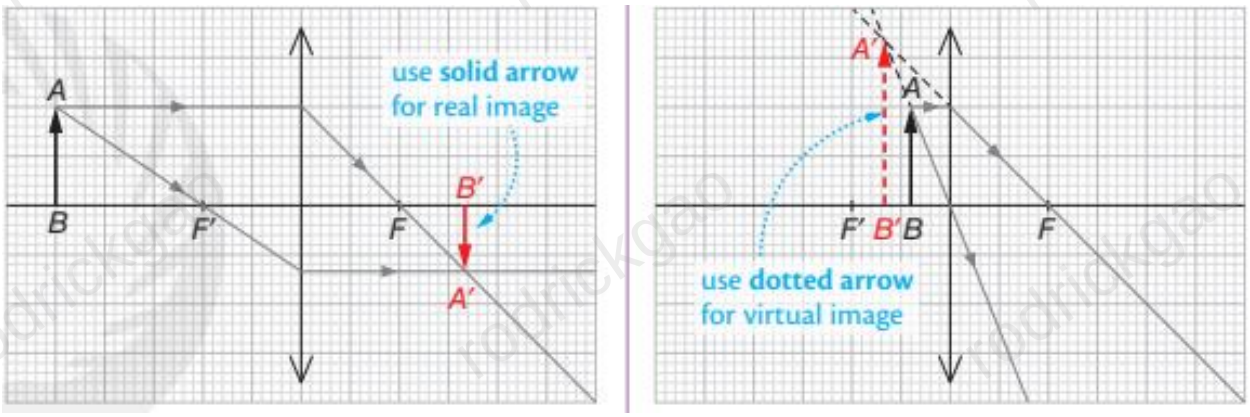
For a real object, a convex lens forms an inverted real image beyond the focal distance and an upright enlarged virtual image within it. At the focal point, emerging rays are parallel and no finite-distance image forms. Real images are diminished for objects beyond twice the focal length, equal-sized at twice it, and enlarged between one and two focal lengths.

3.3.4 Linear magnification

Linear magnification compares image height with object height and can also be obtained from the magnitudes of image distance and real-object distance. Values above 1 mean enlargement and below 1 mean reduction. This card uses a non-negative size ratio, so orientation is stated separately. Linear magnification is not area magnification.

$$m = \frac{h_i}{h_o} = \frac{|\,v\,|}{u}$$

Linear magnification as a size ratio is non-negative. h_i, h_o: image and object heights (same units); u > 0: real-object distance; v: signed image distance. The textbook's v/u uses distance magnitudes here; orientation is stated separately.



Left: an object beyond the focus forms an inverted real image where rays converge. Right: an object within the focal distance forms an upright virtual image where backward extensions meet.

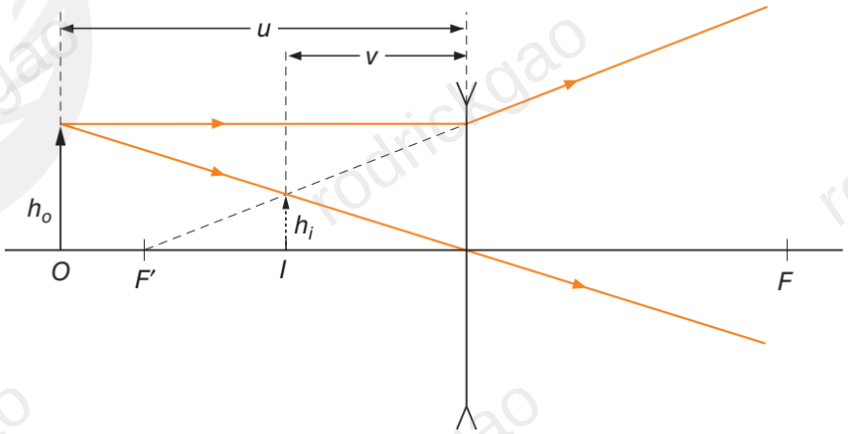


Fig 3.3e *Linear magnification of the image formed by a concave lens.*

Diverging rays trace back to a virtual, upright, smaller image on the object side. The diagram labels distance magnitudes; signed lens calculations use v < 0.

Textbook sources: 3A03 · PDF 13, 19, 20, 30

3.3.5 · 3.3.6 · 3.3.7 Concave lenses and lens calculations

3.3.5 · 3.3.6 Images formed by a concave lens

A concave lens diverges incident light. For a real object in front, backward extensions of emerging rays meet on the object's side, forming an upright, diminished virtual image between the lens and its near focal point. Use a ray parallel to the principal axis and a ray through the optical centre to locate it.

3.3.7 The thin-lens equation and signs

The thin-lens equation relates object distance, image distance and focal length for paraxial rays. Here real-object and real-image distances are positive, virtual-image distance is negative, and focal length is positive for convex and negative for concave lenses. Establish signs and consistent length units before substitution. An infinitely distant object's reciprocal distance is zero.

$$\frac{1}{u}+\frac{1}{v}=\frac{1}{f}$$

Thin lens, paraxial rays. Real object: u > 0; real image: v > 0; virtual image: v < 0. Converging lens: f > 0; diverging lens: f < 0. All distances in the same units.

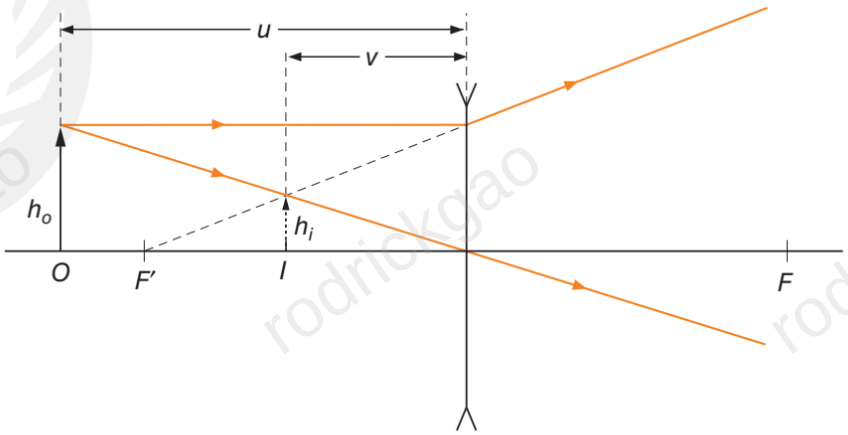
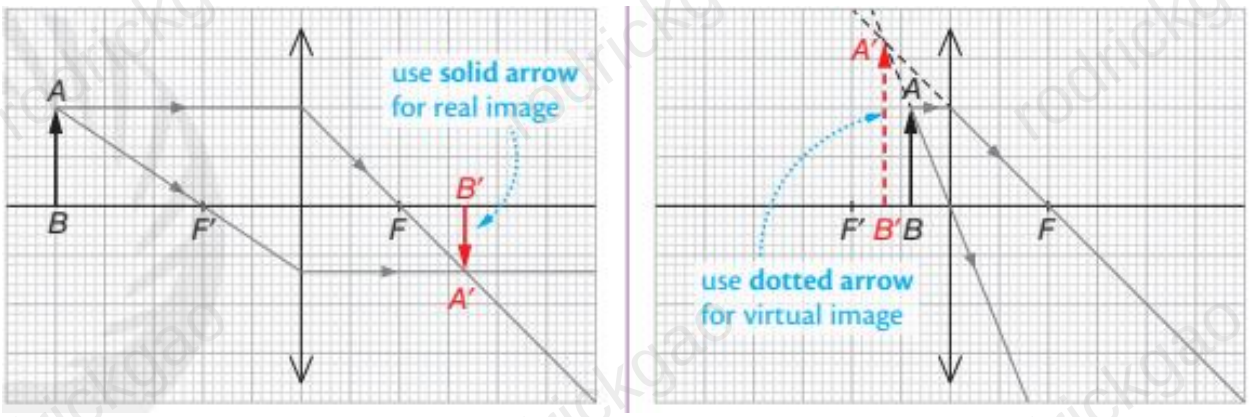


Fig 3.3e *Linear magnification of the image formed by a concave lens.*

Diverging rays trace back to a virtual, upright, smaller image on the object side. The diagram labels distance magnitudes; signed lens calculations use v < 0.



Left: an object beyond the focus forms an inverted real image where rays converge. Right: an object within the focal distance forms an upright virtual image where backward extensions meet.

Textbook sources: 3A03 · PDF 13, 24, 30, 37